

Name	Subject Class	Class	Candidate Number
	2BIX01		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2014

BIOLOGY

HIGHER 1

Paper 2

8875/02
25 AUGUST 2014
2 hours

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.
Fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5 or 6	
Total	60

This question paper consists of **10** printed pages.

[Turn over

- 1 Fig. 1.1 is an electron micrograph of part of a chloroplast in a plant cell.

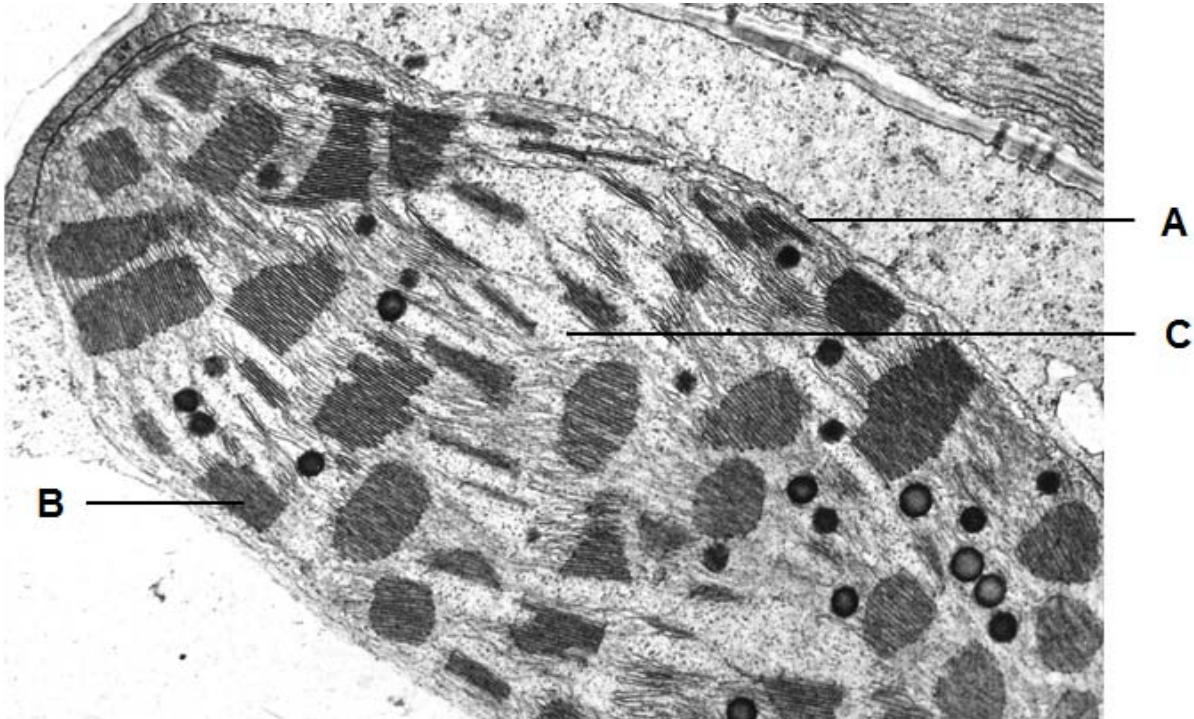


Fig. 1.1

- (a) Explain **two** features of starch that allow it to be stored within the chloroplast.

1. (Hydrophilic) hydroxyl groups all pointing inwards of helix / starch is insoluble in water thus has no effect on osmotic potential of organelle;
2. Amylose helical in shape / amylopectin highly branched, hence can be compacted and stored within a smaller volume of space;

@ 1m, max 2

[2]

- (b) State **one** similarity and **one** difference between structures **A** and **B**.

Similarity:

1. Membrane is made up of two layers of phospholipids;
2. Ref. to amphipathic property of membrane; @ 1m, max 1
3. Ref. to selectively permeable membrane;
4. Ref. to presence of protein carriers;

Difference:

5. Double membrane in A vs single membrane in B;
6. Photosynthetic pigments / ATP synthases / electron carriers found on B but not A; @ 1m, max 1

[2]

(c) Describe how ATP is synthesised in the light dependent reaction.

1. **Electrons (of chlorophyll A) are excited/ energised/ photosystems are photoactivated;**
 2. **Transfer of electrons along the ETC (in the order of decreasing energy level), energy released is harnessed to pump protons from stroma into thylakoid space;**
 3. **Ref. to source of H^+ is also from photolysis of water;**
 4. **Proton pool / proton reservoir is generated;**
 5. **Diffusion of protons down the electrochemical gradient/ ref. to proton motive force/ proton gradient/ concentration gradient via ATP synthase for ATP synthesis (by chemiosmosis);**
- @ 1m**

[3]

(d) The enzyme Rubisco is found in **C**. Suggest and explain the effect on the rate of light-dependent reactions if the gene coding for the enzyme Rubisco is mutated.

1. ***Rate of light-dependent reactions decrease;**
2. **Mutation of gene results in a change in 3D conformation in Rubisco and hence it becomes non-functional / Rubisco is not produced;**
3. **No carbon fixation/ no formation of glycerate-3-phosphate (GP), hence GP no longer reduced to glyceraldehyde-3-phosphate;**
4. **No regeneration of $NADP^+$ and ADP for light-dependent reactions to take place;**

@ 1m, * compulsory point

[4]

[Total: 10]

2 Fig. 2.1 shows the structure of molecules and some associated processes found in a cell.

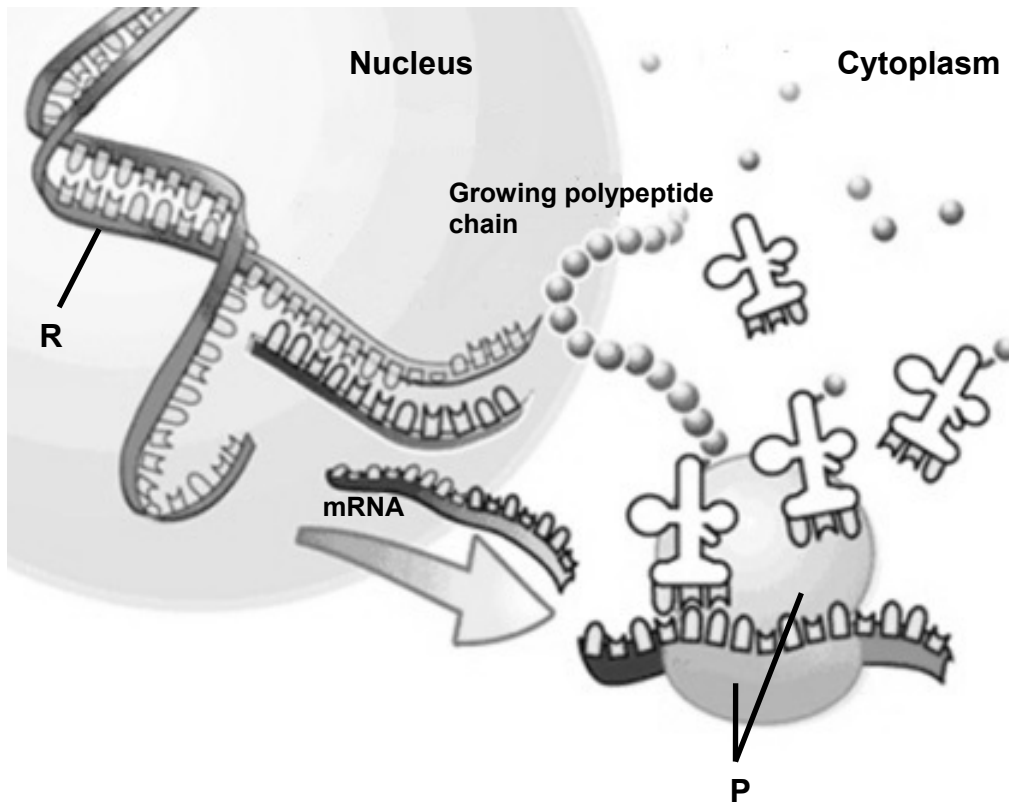


Fig. 2.1

(a) (i) Name the type of bond holding the two strands of **R** together.

Hydrogen bonds; [1]

(ii) Describe the importance of this type of bond named in (a)(i) in **R**.

1. **Allows for semi-conservative replication / transcription via complementary base pairing;**
2. **Allows for DNA repair via complementary base pairing;**
3. **Hydrogen bonds help to stabilize R;**
4. **Hydrogen bonds can be easily broken by RNA polymerase/ helicase to unzip the DNA molecule;**

@ 1m, max 2 [2]

(b) State **one** role of **P** in translation.

1. **Aligns codons in mRNA in ribosome to (base) pair with anticodons of amino-acyl tRNA;**
2. **Orientates a large variety of enzymes (e.g. peptidyl transferase) in correct spatial arrangement;**
3. **Catalyses formation of peptide bonds between (adjacent) amino acids;**

@ 1m, max 1 [1]

Not all mRNAs which enter the cytoplasm are used immediately for translation. Initiation of translation is prevented because some mRNAs lack poly-A tails of sufficient length. At the appropriate time, a polyadenylation enzyme will add more adenine nucleotides to lengthen the poly-A tail, allowing translation of these mRNAs to begin.

(c) Describe the mode of action of the polyadenylation enzyme.

1. **The polyadenylation enzyme lowers activation energy;**
2. **For the free adenine nucleotides to be brought into close proximity with the mRNA / orientation surface / strain effect / microenvironment effect / acid-base catalysis;**
3. **3D shape of active site of enzyme is complementary to that of adenine nucleotides, thus forming enzyme-substrate complexes;**
4. **With ref. to change in 3D conformation for a better fit/ tighter binding between substrate and enzyme / Ref. to lock and key hypothesis;**
5. **Resulting in formation of phosphodiester bonds between the adenine nucleotides;**

@ 1m, max 4

[4]

In the cell, free adenine nucleotides that are added to the poly-A tail exist as adenosine triphosphate. A molecule of adenosine triphosphate is shown in Fig. 2.2a below.

Introduction of Cordycepin into the cell prevents the polyadenylation enzyme from carrying out its function at the 3' end of mRNA. Translation does not begin in the presence of Cordycepin. Fig. 2.2b shows the structure of Cordycepin.

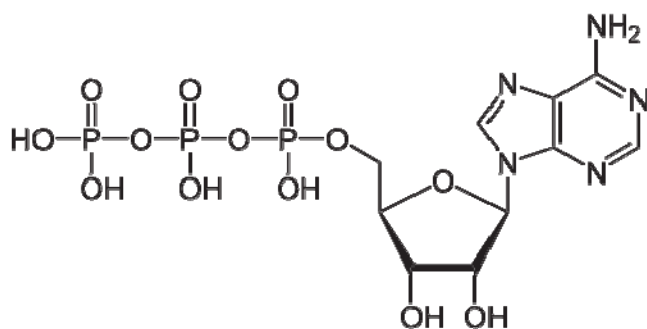


Fig. 2.2a

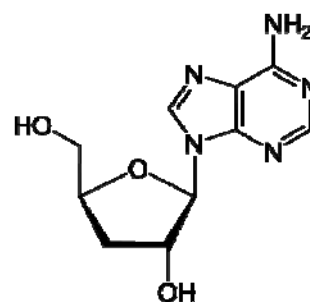


Fig. 2.2b

(d) Suggest how Cordycepin prevents translation of mRNA in the cytoplasm.

1. **Cordycepin is a competitive inhibitor of the polyadenylation enzyme / 3D structural resemblance of inhibitor to that of substrate;**
2. **As enzyme is unable to add more adenine nucleotides to the poly-A tail,**
3. ***Poly-A tail is not of the sufficient length to initiate translation;**

@1m, *compulsory point

@ 1m [2]

[Total: 10]

[Turn over

- 3 In a species of bird, feathers may be white, black or black with white spots (spotted). In this same species, beaks may be long or short.

In a study to check on genetic diversity of this species of bird, a research ecologist examined the results obtained from the crosses of two pairs of birds. The results of the crosses are shown in Table 3.1 below.

Table 3.1

	Cross	Ratio of F1 phenotypes
1	Black bird with long beak crossed with spotted bird with long beak	3 black with long beak : 3 spotted with long beak : 1 black with short beak : 1 spotted with short beak
2	Spotted bird with long beak crossed with white bird with short beak	1 spotted with long beak: 1 white with long beak

- (a) Using a genetic diagram, explain the results of the cross between a black bird with long beak and a spotted bird with long beak as shown in Table 3.1.

Let C^B represents allele for black feather and C^W represents allele for white feather
L represent dominant allele for long beak, and l represent recessive allele for short beak

Phenotype of parents: Black with long beak x Spotted with long beak

Genotype of parents: $C^B C^B L l$ x $C^B C^W L l$;

Gametes: $(C^B L) (C^B l)$ $(C^B L) (C^B l) (C^W L) (C^W l)$;

F1 genotypes: Punnett Square

Gametes	$C^B L$	$C^B l$	$C^W L$	$C^W l$
$C^B L$	$C^B C^B L L$ Black, long beak	$C^B C^B L l$ Black, long beak	$C^B C^W L L$ Spotted, long beak	$C^B C^W L l$ Spotted, long beak
$C^B l$	$C^B C^B l l$ Black, long beak	$C^B C^B l l$ Black, short beak	$C^B C^W l l$ Spotted, long beak	$C^B C^W l l$ Spotted, short beak

F1 Phenotypic ratio:

3 black with long beak : 3 spotted with long beak : 1 black with short beak : 1 spotted with short beak

- 1 m for correct genotype for parents (mk given only when key is present);
- 1 m for correct gametes;
- 1 m for correct genotype of offspring in Punnett square;
- 1 m for correct phenotype of offspring in Punnett square;

- (b) Describe the cross you would carry out to determine if a randomly picked F_1 offspring with black feathers and long beak from the cross in part (a) is homozygous for beak length.

1. **Cross the F_1 offspring with a black / white bird with short beak;**
2. **Presence of offspring with long and short beaks indicate heterozygosity/ All offspring with long beak indicate homozygosity;**

@ 1m [2]

In order to confirm the genotype of the above F_1 offspring, polymerase chain reaction (PCR) was carried out on a DNA sample of the bird. The investigator added a mixture to the DNA sample and in a few hours, the tiny sample of DNA was amplified millions of times.

- (c) Identify the components in the mixture that was added with the DNA sample and their respective functions for PCR to amplify the sample of DNA present.

1. **Both forward and reverse (DNA) Primers to flank targeted DNA region to be amplified;**
2. **Deoxyribonucleotides (with all four types of bases) for replication of targeted DNA region by complementary base pairing;**
3. **Taq polymerase to read and replicate DNA template strands by formation of phosphodiester bond (between deoxyribonucleotide and OH gp at the 3' end);**

@ 1m [3]

- (d) Explain **one** possible limitation of this procedure.

1. **Need to know the nucleotide sequence on each side of the region of DNA to be amplified in order to synthesise the PCR primers;**
2. **Errors in base sequence of amplified DNA copies due to the absence of built-in proofreading in Taq polymerase;**

@ 1m

[Total: 10]

- 4 Primate species have the potential to visually perceive three colour spectrums – blue, green and red.

Fig. 4.1 shows a phylogenetic tree of primate evolution. The common ancestor for all primate species was thought to be dichromatic, being able to perceive blue and one other colour only (hence unable to distinguish green and red colour). Only the species within the group of “apes” and “old world monkeys” are found to be consistently trichromatic and are able to perceive all three colours.

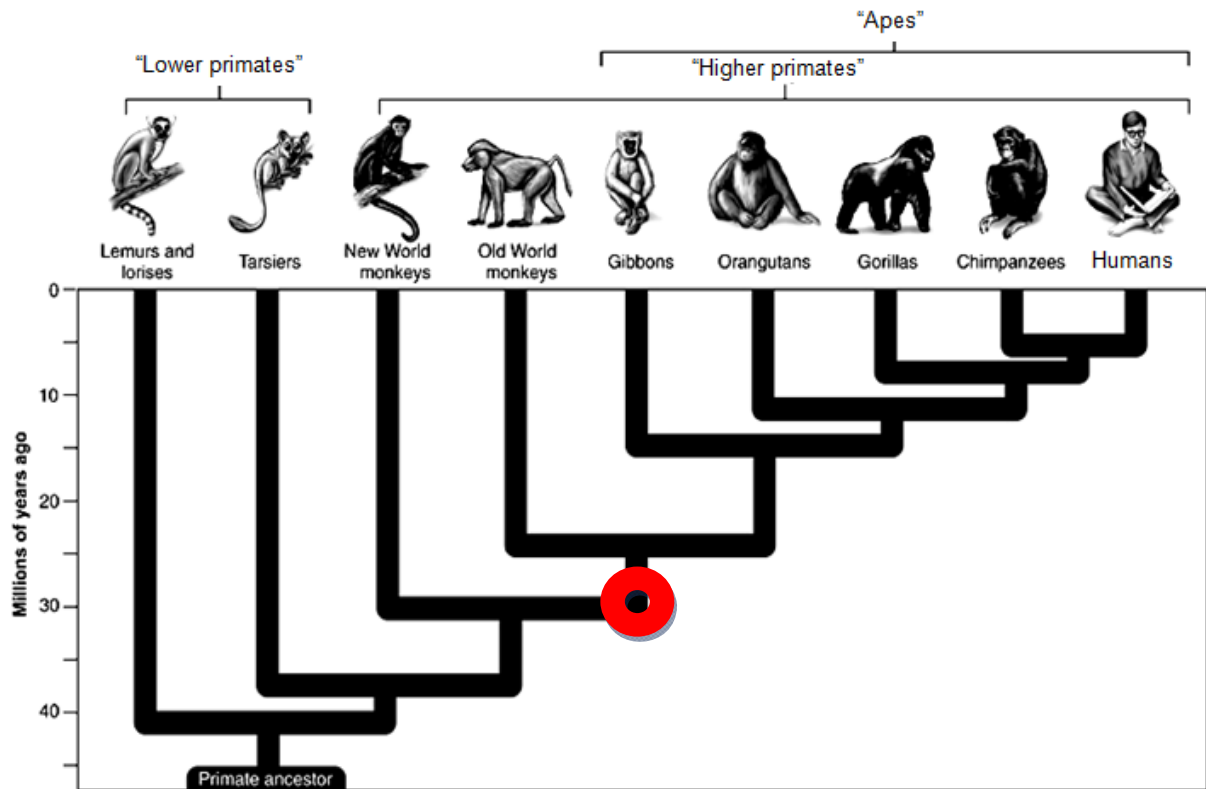


Fig. 4.1

- (a) (i) Using an arrow, indicate on Fig. 4.1 the point where trichromatic vision is likely to have evolved in primates. [1]
- (ii) Suggest how trichromatic vision may be of survival value to fruit-feeding primates in forest habitats.

Allow primates to distinguish (ripened) fruits from green background;

@ 1m

[1]

(ii) Explain how natural selection has resulted in the evolution of trichromatic vision in a population.

1. **Individuals with this trait would have a selective advantage / be selected for (because they are better able to obtain food);**
 2. **They would survive to produce viable offspring / experience higher reproductive success / have greater fitness;**
 3. **Genes coding for the trait will be passed down to subsequent generations;**
***Over time, this leads to an increase in frequency of the trait in the**
 4. **population;**
- @ 1m, * compulsory point**

[3]

Almost all known species of "new world monkeys" have dichromatic vision. The gene for blue colour vision (gene B) is found on chromosome 7, while green colour vision and red colour vision are controlled by different alleles of the same gene (gene G/R) found on chromosome X.

Howler monkeys (*Alouatta* spp.) are a genus of "new world monkeys" with trichromatic vision. They share the same gene B with other species of "new world monkeys". However, on chromosome X, the gene for green colour vision (gene G) is found on a different locus than the gene for red colour vision (gene R). Both gene G and gene R may therefore coexist on a single chromosome.

(b) Describe the mutation that could have led to trichromatic vision in howler monkeys.

1. **Individuals with this trait would have a selective advantage / be selected for (because they are better able to obtain food);**
2. **More survive to reproduce / experience higher reproductive success / have greater fitness;**
3. **Genes coding for the trait will be passed down to subsequent generations;**
4. ***Over time, this leads to an increase in allele frequency for the trait in the population;**

@ 1m, pt 4 compulsory, max 3

[3]

Cone cells are somatic cells found in the retina of the eye which are responsible for colour vision.

(c) (i) Describe **two** differences between cone cells and adult stem cells.

1. **Differentiated vs undifferentiated / unspecialised;**
2. **Not capable of continued cell division vs capable of self-renewal for long periods;**
3. **Cannot further differentiate vs can further differentiate to specialised cell types under appropriate chemical signals;**

@ 1m, max 2

[2]

(ii) State the function of adult stem cells in the body.

Generate replacements for cells lost through injury / wear and tear / AVP;

@ 1m

[1]

[Total: 10]

Section B

5 (a) Describe the structure and functions of lysosome.

[6]

Structural Features	Function
1. Membrane bound organelle / ref to phospholipid bilayer;	2. To maintain different chemical environment (acidic environment); 3. To prevent hydrolytic enzyme from interfering with other processes in the cytoplasm;
4. Allow high concentration of enzymes to accumulate;	5. for high rate of hydrolysis;
6. Presence of proton pump embedded on phospholipid bilayer of lysosome;	7. Active transport of H ⁺ into lysosome; 8. Create acidic environment for hydrolytic enzymes / optimal pH of hydrolytic enzyme is acidic;
9. Presence of hydrolytic enzymes in lysosome;	10. For digestion of ingested particles 11. For break down of defunct or surplus organelle; 12. For <u>autolysis</u> of cell;
13. Fluidity of membrane;	14. Allow lysosomal membrane to fuse with endocytic /phagocytic vesicle; 15. Rupture of membrane to release hydrolytic enzymes;
16. modification of molecules in membrane;	17. prevent digestion of membrane;
18. presence of carrier/ transporter on lysosome membrane;	19. transport products of digestions back into cytosol;

@1m, max 3 for structure, max 3 for function

(b) Describe how variations may arise in a population and explain their role in evolution. [8]

- 1 Crossing over in prophase I of meiosis I;
- 2 Independent assortment in metaphase I and anaphase I / metaphase II and anaphase II or meiosis;
- 3 Random fusion of gametes during fertilization;
- 4 Leads to formation of new combination of alleles (in gametes);
- 5 (Gene and chromosomal) mutations;
- 6 Leads to formation of new alleles;
- 7 Genetic variation contributes towards phenotypic variation;
- 8 Natural selection acts on variation in a population; OWTTE
- 9 Where phenotypes that are best adapted to the environment will be selected for; / Without variation in a population, selection pressure acts equally on all individuals;
- 10 Hence different individuals experience differential reproductive success; / Individuals will experience the same reproductive success;
- 11 Leading to a change in allele frequency in a population over time; / There will be no change in allele frequency in a population over time;

- (c) With reference to Bt. corn, explain the significance of genetic engineering to solve the global demand for food. [6]

- 1 Genetic engineering: process of changing the genetic make-up of a living organism or cell by introduction of gene from another species or mutation into a specific gene;
- 2 Bt genes (from *Bacillus thuringiensis*) is inserted into corn crops (via use of Ti plasmid of *Agrobacterium tumefaciens*);
- 3 Bt toxin/protein produced kills the European corn borer due to damage caused to the gut wall;
- 4 Thus, less crops are eaten and yield is increased;
- 5 Use of Bt corn reduces the need for pesticide applications, hence reducing cost, labour;
- 6 Ref. to greater supply of food;
- 7 Improve livelihood and alleviate poverty;
- 8 There are no issues of ingestion of insecticides by humans, which may lead to poisoning or other harmful effects; @ 1m

[20]

- 6 (a) Compare and contrast the structures of glycogen and a globular protein. [8]

Similarities:

1. Both are polymers;
2. Both have hydrogen bonds;
3. Both have helical structures;

Differences:

Features	Glycogen	Globular protein
Elements	4. C, O, H	C, O, H, N, S
Monomer	5. α -glucose	amino acid;
Bond between monomers	6. glycosidic bonds	peptide bonds;
Other types of bonds	7. Hydrogen bonds	hydrogen bonds, disulphide bonds, ionic bonds, hydrogen bonds;
Branching	8. Yes	No;
Crosslinks	9. No	Yes;
Orientation of hydroxyl groups	10. point inwards	Ref. to the exterior;

@ 1m, max 8

- (b) Explain the production of a small yield of ATP from anaerobic respiration in mammalian cells. [6]

1. Ref. to incomplete/partial oxidation of glucose during glycolysis in the absence of oxygen;
2. 1 molecule of glucose is broken down to 2 molecules of pyruvate;
3. NAD^+ is required to oxidise glucose;
4. With the net yield of 2 ATP and 2 reduced NAD;
5. Ref. to ATP production by substrate level phosphorylation;
6. Pyruvate will be reduced to lactate by accepting H atoms from reduced NAD;
7. Ref. to lactate dehydrogenase;
8. NAD^+ is regenerated so that glycolysis can continue to occur (to yield small amount of ATP);
@ 1m

(c) Outline the procedure for cloning a eukaryotic gene with the use of bacterial plasmid.

[6]

1. Extract mRNA of gene of interest from eukaryotic cells;
2. Synthesize ds-cDNA using reverse transcriptase (and RNase);
3. Add linkers (with appropriate restriction site) to cDNA;
OR
4. Isolate gene of interest from eukaryotic cell;
5. Digest gene of interest/cDNA and plasmid with same restriction enzyme to produce complementary sticky ends;
6. Allow DNA and cut plasmids to anneal by formation of H-bonds between complementary bases at sticky ends;
7. Ref. to added DNA ligase for phospho-diester bonds formation;
8. before mixing with bacteria for transformation via heat shock/electroporation;
9. Selection of recombinant bacteria on nutrient medium with appropriate chemicals;
10. Culture selected bacteria in nutrient medium;
@ 1m, max 6

[20]